

MedGraphRAG: A Hybrid Dense–Sparse–Graph Retrieval Architecture for Evidence-Grounded Clinical Question Answering in Medical Oncology

Abstract

Large language models (LLMs) are increasingly explored as clinical information-retrieval tools, yet their tendency to hallucinate --- generating fluent but unsupported claims --- remains a critical barrier to safe deployment in oncology and other high-stakes medical domains. Standard Retrieval-Augmented Generation (RAG) mitigates parametric hallucination by conditioning generation on retrieved evidence, but single-channel retrieval (typically dense embedding search alone) exhibits systematic blind spots: it under-retrieves rare, lexically distinctive clinical terms and cannot surface information reachable only through multi-hop entity relationships. This paper presents MedGraphRAG, a tri-modal retrieval architecture that fuses dense semantic search (**BAAI/bge-base-en-v1.5** over ChromaDB), sparse lexical search (BM25Okapi), and multi-hop knowledge-graph traversal (SciSpaCy biomedical named-entity recognition over a NetworkX graph, scored with an Inverse-Entity-Frequency topological decay function) into a single fused, cross-encoder-reranked (**BAAI/bge-reranker-base**) retrieval pipeline. Generated answers are further verified through sentence-level Natural Language Inference (NLI) entailment checking against cited evidence, with mandatory per-claim citation attribution. We evaluate MedGraphRAG on a 200-question oncology gold-standard benchmark and report a controlled, five-condition ablation study (Baseline, No-Graph, No-BM25, No-Reranker, Dense-Only) over 500 total inferences, using paired Wilcoxon signed-rank significance testing. Results show that no single retrieval channel is sufficient in isolation: removing the cross-encoder reranker reduces Precision@5 from 0.895 to 0.328 (\$p<0.001\$), removing BM25 reduces Groundedness from 0.912 to 0.638 (\$p<0.01\$), and a dense-only configuration exhibits significantly elevated hallucination (\$0.092 \rightarrow 0.391\$) despite being ~43% faster. These findings provide statistically grounded evidence for hybrid, graph-augmented retrieval as a hallucination-mitigation strategy in clinical question answering.

Benchmark Results (Synchronized)

Condition	Retrieval Acc.	Precision@5	Recall@5	Faithfulness	Answer Relevance	Groundedness	Hallucination Rate (↓)	Latency (s)
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Baseline (Full)	0.930 ± 0.250	0.895 ± 0.034	0.978 ± 0.053	0.908 ± 0.028	0.915 ± 0.020	0.912 ± 0.037	0.092 ± 0.028	25.572 ± 9.812
No_Graph	0.920 ± 0.271	0.464 ± 0.185	0.970 ± 0.060	0.696 ± 0.065	0.843 ± 0.048	0.755 ± 0.397	0.304 ± 0.065	25.402 ± 11.581
No_BM25	0.860 ± 0.347	0.408 ± 0.215	0.960 ± 0.066	0.674 ± 0.085	0.841 ± 0.048	0.638 ± 0.454	0.326 ± 0.085	31.473 ± 9.185
No_Reranker	0.850 ± 0.357	0.328 ± 0.207	0.972 ± 0.059	0.684 ± 0.082	0.836 ± 0.048	0.684 ± 0.444	0.316 ± 0.082	18.137 ± 4.813
Dense_Only	0.800 ± 0.400	0.410 ± 0.244	0.951 ± 0.070	0.609 ± 0.243	0.734 ± 0.288	0.672 ± 0.448	0.391 ± 0.243	14.217 ± 6.236